

Sequential Logic Circuits - I

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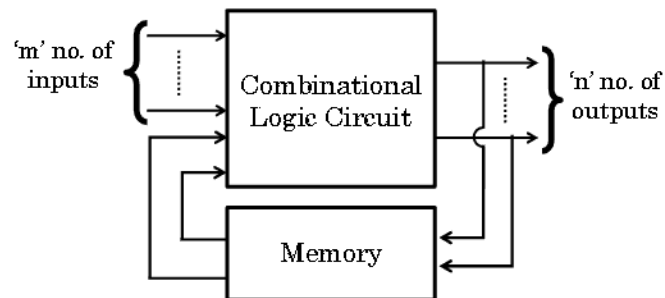
Agenda

- ◆ Sequential Logic circuits
- ◆ Latches
- ◆ Flipflops
- ◆ Delays in Sequential Circuits

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Sequential Logic Circuits

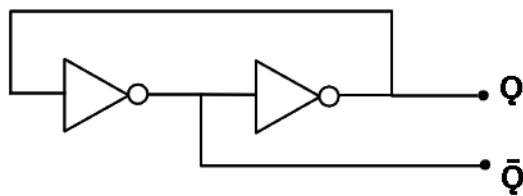


- ♦ The output depends on the input & past state of the circuit.
- ♦ Memory Units

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Memory units



Q_n	Q_{n+1}
1	1
0	0

The above circuit is capable of restoring the data which is available at instant of power ON condition.

This indicates that cascaded combination of 2 inverters with a feedback connection is capable of storing 1 bit information.

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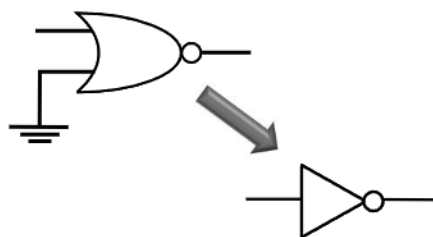
Logic Circuits as memory units

- ♦ The cascade connection of 2 inverters with a feedback retains the past information
- ♦ And hence it can be considered as a 1-bit memory element or 1-bit memory cell.
- ♦ But, there is no input port and hence we don't have the control over the data that has to be stored.
- ♦ Let's try to modify this circuit and provide i/p ports by which we can customize the data that is going to be stored in the memory cell.

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1-bit memory cell



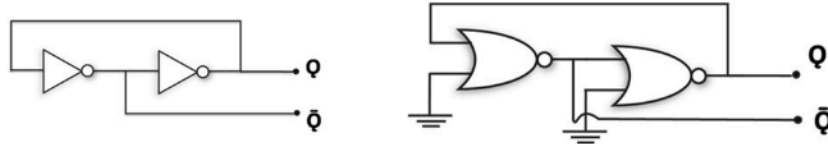
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

- ♦ If $A = 0$, then $Y = B'$.
- ♦ This states that if one of the input to NOR gate is grounded, then that NOR gate will act as a NOT gate.

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1-bit memory cell

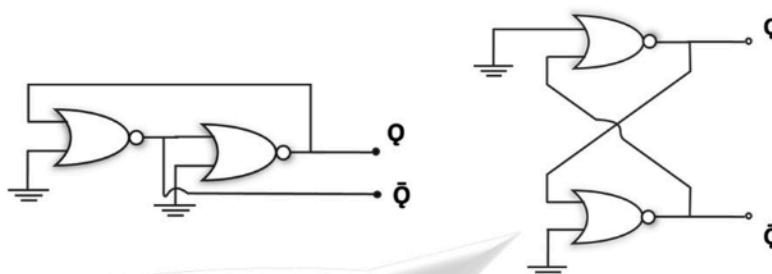


- ♦ In this memory cell, these two inverters can be replaced with a pair of NOR gates with one input connected to ground.
- ♦ In this circuit, the feedback is in such a way that, output of first inverter is input to the second inverter & output of second inverter is the input to the first inverter.

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1-bit memory cell

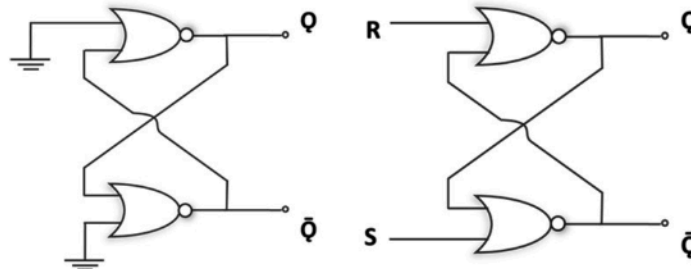


But, even with modification, there is no input port and hence we don't have the control over the data to be stored.

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1-bit memory cell

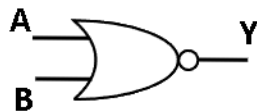


- ♦ This is how, RS Latch has been evolved.
- ♦ Latch is nothing but a one bit memory cell which is capable of retaining the past state output.
- ♦ Since, latch is evolved from a cascaded combination of a pair of inverters, the two outputs of a latch should be always complement to each other.

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Behaviour of a NOR gate



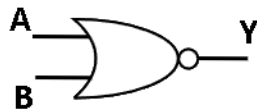
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

- ♦ This states that if one of the input to NOR gate is 0, then the output of the NOR gate will be complement to the remaining input.
- ♦ This indicates that, if one of the input is 0, then we cannot predict the output of the NOR gate without having the knowledge of the remaining input.

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Behaviour of a NOR gate



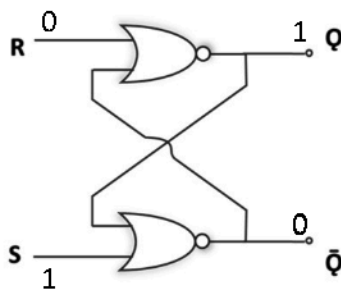
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

- ♦ This states that if one of the input to NOR gate is 1, then the output of the NOR gate will be always 0 irrespective of the remaining input.
- ♦ This indicates that, if one of the input is 1, then we can easily predict the output of the NOR gate without having the knowledge of the remaining input.

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RS Latch



R	S	Q	Q'	State
0	1	1	0	Set

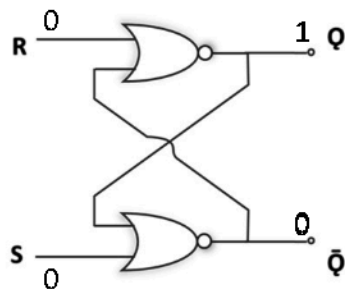
Recollect the fact that, if one of the input to NOR is 1, then output will be '0'.

The outputs for this input combination is 1 & 0.

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RS Latch



R	S	Q	Q'	State
0	1	1	0	Set
0	0	1	0	No Change

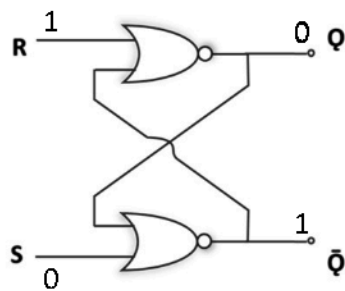
In this case also, the outputs for this input combination is 1 & 0.

Even though, there is some change input, the output is same as the output in the past state

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RS Latch



R	S	Q	Q'	State
0	1	1	0	Set
0	0	1	0	No Change
1	0	0	1	Reset

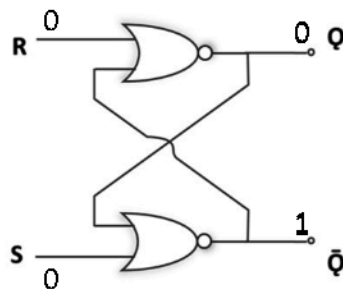
Recollect the fact that, if one of the input to NOR is 1, then output will be '0'.

The outputs for this input combination is 0 & 1.

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RS Latch



R	S	Q	Q'	State
0	1	1	0	Set
0	0	1	0	No Change
1	0	0	1	Reset
0	0	0	1	No Change

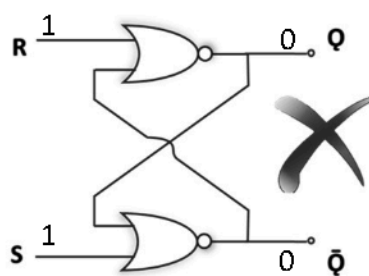
In this case also, the outputs for this input combination is 0 & 1.

Even though, there is some change in the input, the output is same as the output in the past state

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RS Latch



R	S	Q	Q'	State
0	1	1	0	Set
0	0	1	0	No Change
1	0	0	1	Reset
0	0	0	1	No Change
1	1	0	0	Invalid

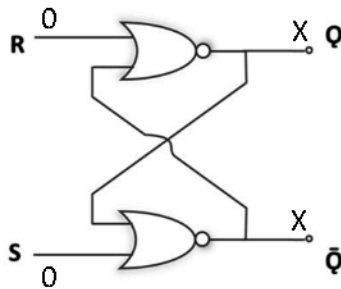
Observe what happens if invalid combination of i/p is followed by no change combination of input.

Both the outputs of a latch should be complement to each other. But, in this case, both the outputs are same.

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RS Latch



R	S	Q	Q'	State
0	1	1	0	Set
0	0	1	0	No Change
1	0	0	1	Reset
0	0	0	1	No Change
1	1	0	0	Invalid
0	0	X	X	RACE

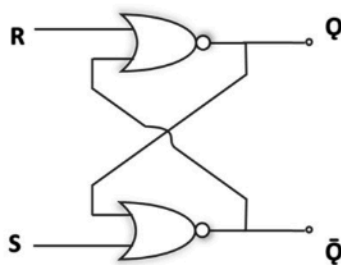
Q	1	0	1	0	1
Q'	1	0	1	0	1

1. At the o/p terminals, both 1 & 0 will be racing with each other even though there is no change in the i/p .
2. If invalid combination of i/p is followed by no change combination of i/p, race around condition occurs.

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RS Latch



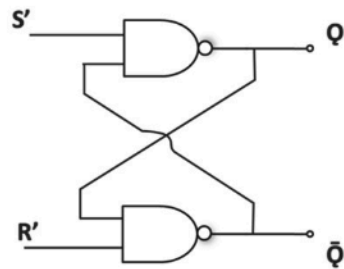
R	S	Q	Q'	State
0	1	1	0	Set
0	0	1	0	No Change
1	0	0	1	Reset
0	0	0	1	No Change
1	1	0	0	Invalid
0	0	-	-	RACE

1. Since, the Set input is HIGH, output is SET.
2. Since, the Reset input is HIGH, output is RESET.
3. That's why, RS Latch with NOR Gates is said to be Active HIGH Latch.

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RS Latch – NAND Gates



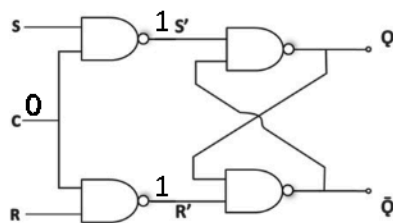
S'	R'	Q	Q'	State
0	1	1	0	Set
1	1	1	0	No change
1	0	0	1	Reset
1	1	0	1	No change
0	0	1	1	Invalid
1	1	-	-	RACE

- ♦ The only difference is RS Latch with NOR gates is active HIGH Latch where as RS Latch with NAND gates is active LOW Latch.
- ♦ In this case, (0 & 0) input combination is an invalid combination of input and (1 & 1) input combination doesn't bring any change in the output.

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Gated RS Latch

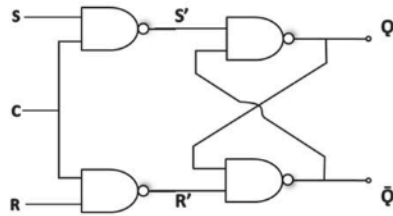


C	S	R	S'	R'	Q	Q'	State
0	X	X	1	1			No Change
1	0	0	1	1			No Change
1	0	1	1	0	0	1	Reset
1	1	0	0	1	1	0	Set
1	1	1	0	0	-	-	Invalid

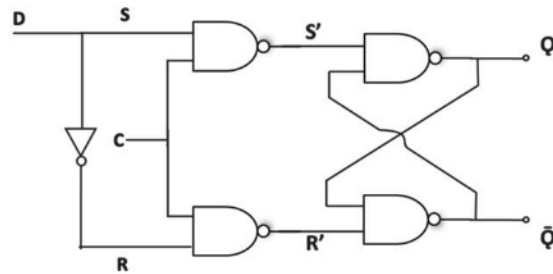
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Gated D Latch



C	S	R	S'	R'	Q	Q'	State
0	X	X	1	1			No Change
1	0	0	1	1			No Change
1	0	1	1	0	0	1	Reset
1	1	0	0	1	1	0	Set
1	1	1	0	0	-	-	Invalid



C	D	Q	Q'	State
0	X			No Change
1	0	0	1	Reset
1	1	1	0	Set

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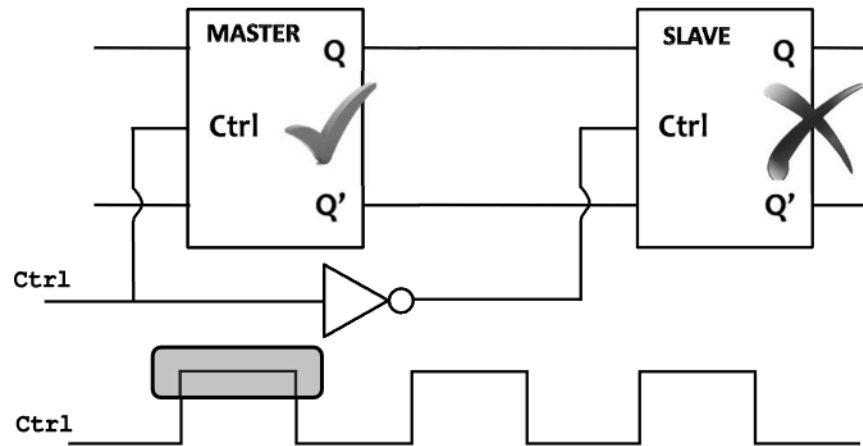
Gated D Latch

- ◆ In Controlled RS Latch, RACE Condition will occur if both inputs are 1.
- ◆ In controlled RS Latch, if both the inputs are complement to each other, the output follows the S input.
- ◆ So, to avoid RACE conditions, both the inputs must be complement to each other.
- ◆ This is how D Latch is constructed.
- ◆ Output Q follows the input D, said to be transparent Latch.

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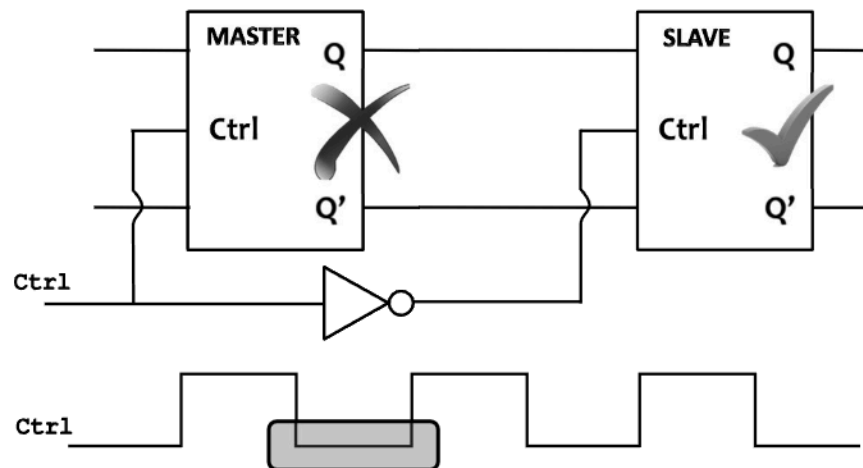
Master Slave Configuration



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Master Slave Configuration



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Master Slave Configuration

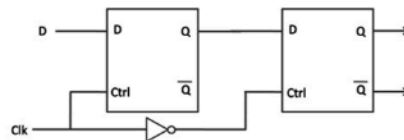
- ♦ The output at the final stage will be depending on the input available at the instant of negative edge.
- ♦ Master Slave configuration of two latches is treated as a flip flop.

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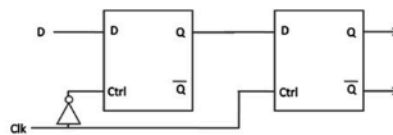
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Construction of Flipflops

- ♦ Negative Edge Sensitive flipflop is constructed by using HIGH level latch as master and LOW level latch as slave.



- ♦ Positive Edge Sensitive flipflop is constructed by using LOW level latch as master and HIGH level latch as slave.



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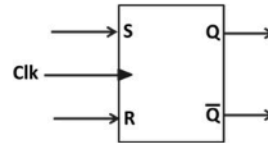
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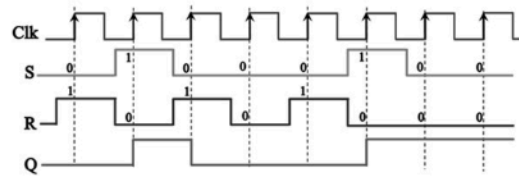
SR Flip-flop

Truth Table

Clk	S	R	Q ₊	State
	X	X	Q	No Change
	0	0	Q	No Change
	0	1	0	Reset
	1	0	1	Set
	1	1	-	Invalid



Wave forms



Excitation Table

Q	Q ₊	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

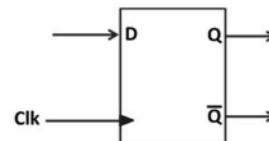
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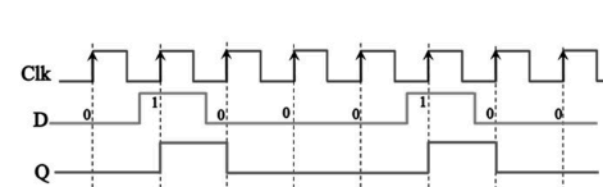
D Flip-flop

Truth Table

Clk	D	Q ₊	State
	X	Q	No Change
	0	0	Reset
	1	1	Set



Wave forms



Excitation Table

Q	Q ₊	D
0	0	0
0	1	1
1	0	0
1	1	1

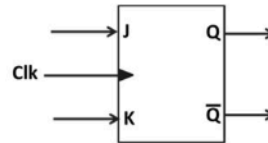
174

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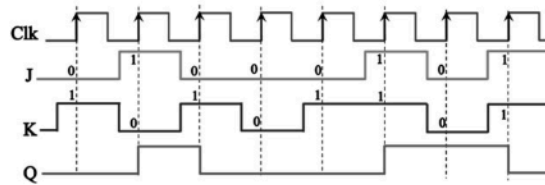
JK Flip-flop

Truth Table

Clk	J	K	Q_+	State
	X	X	Q	No Change
	0	0	Q	No Change
	0	1	0	Reset
	1	0	1	Set
	1	1	Q'	Toggle



Wave forms



Excitation Table

Q	Q_+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

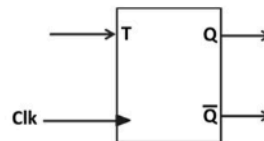
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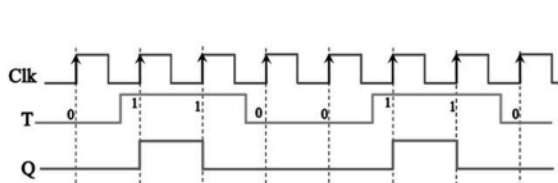
T Flip-flop

Truth Table

Clk	T	Q_+	State
	X	X	No Change
	0	Q	No Change
	1	Q'	Toggle



Wave forms



Excitation Table

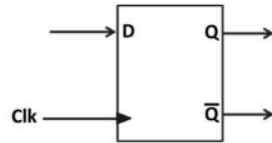
Q	Q_+	T
0	0	0
0	1	1
1	0	1
1	1	0

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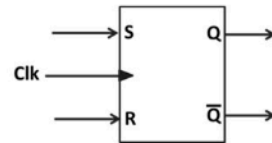
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Symbols of flipflops

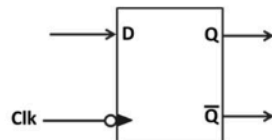
Positive Edge D flipflop



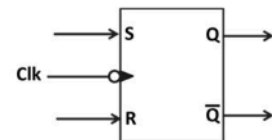
Positive Edge RS flipflop



Negative Edge D flipflop



Negative Edge RS flipflop

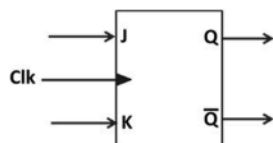


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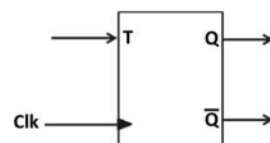
177

Symbols of flipflops

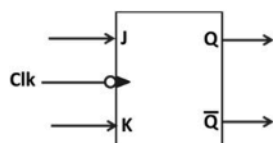
Positive Edge JK flipflop



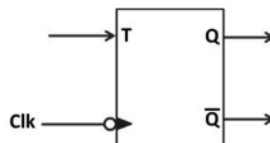
Positive Edge T flipflop



Negative Edge JK flipflop



Negative Edge T flipflop

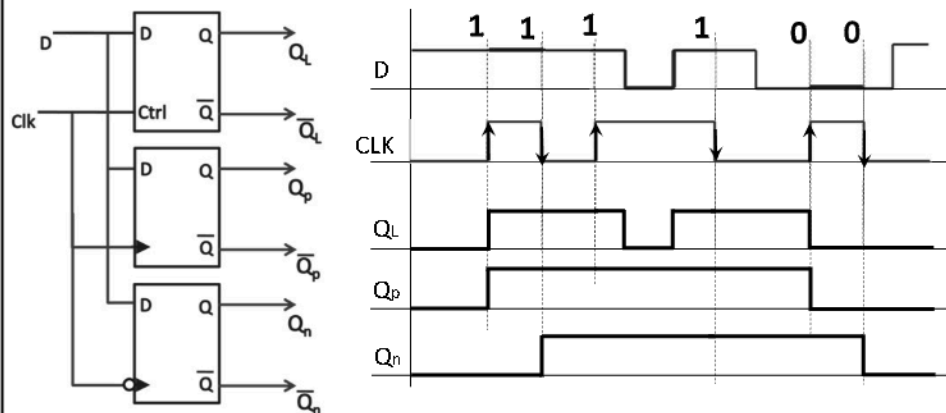


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Timing Waveform

Draw the timing waveform for the below components, which receives same Clk and D input signals.



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Flipflops with Asynchronous inputs.

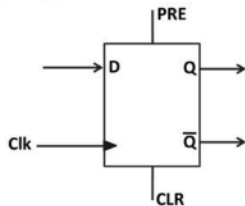
- ♦ The inputs of the flipflops like R, S, D, J, K & T are called synchronous inputs because, these inputs can bring a change in the outputs of the flipflops only at the instant of posedge or negedge of the clock.
- ♦ So, we need to wait till the instant of posedge or negedge of the clock to set or reset the flipflop.
- ♦ Hence, flipflops are provided with a pair of asynchronous inputs (preset & clear) to make the flipflop set or reset at any instant of time without waiting for the active edge of the clock signal.

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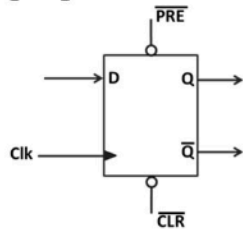
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Flipflops with Asynchronous inputs.

Flipflop with active HIGH Asynchronous Inputs.



Flipflop with active LOW Asynchronous Inputs.



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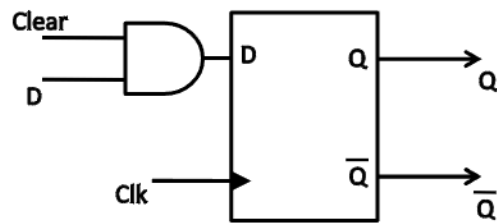
Active HIGH Asynchronous inputs.

- ♦ When preset is HIGH, the flipflop will be driven into set state without waiting for the posedge of the flipflop.
- ♦ When clear is HIGH, the flipflop will be driven into reset state without waiting for the posedge of the flipflop.
- ♦ For normal operation both the asynchronous inputs must be connected to '0'

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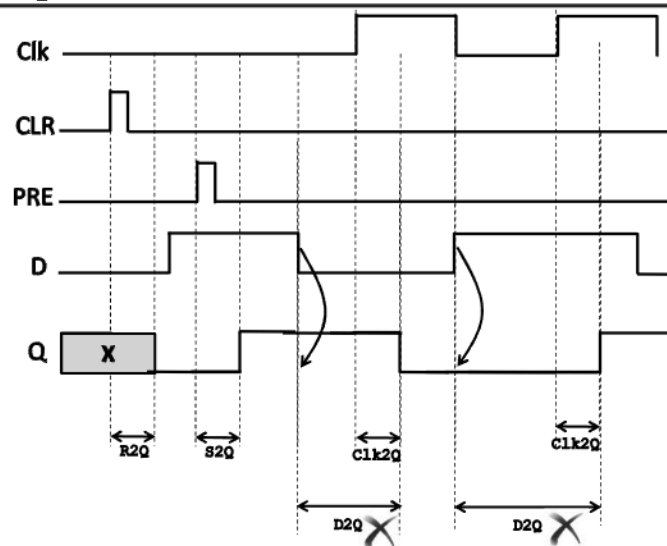
Synchronous Clear



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Delays in Sequential Circuits



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Delays in Sequential Circuits

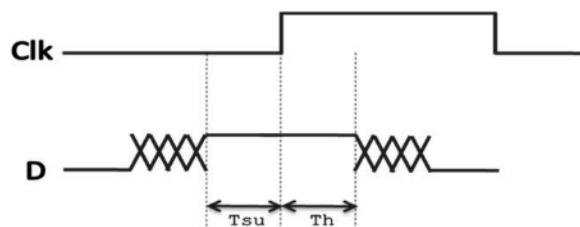
- ♦ **D2Q Delay** : change in input D alone cannot bring a change in the output . So, D2Q delay is not valid.
- ♦ **Clk2Q Delay** : Time taken by the flipflop to provide a stable output after a clock edge occurs.
- ♦ **R2Q Delay** : Time taken by flipflop to come to RESET state after the application of asynchronous CLEAR input.
- ♦ **S2Q Delay** : Time taken by flipflop to come to SET state after the application of asynchronous PRESET input.

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Setup & Hold Times

- ♦ **Setup Time** : is the minimum time interval for which the input signal must be stable (unchanging) prior to the sampling event of the clock for the input signal to be recognized correctly.
- ♦ **Hold Time** : is the minimum time interval for which the input signal must be stable (unchanging) following the sampling event of the clock for the input signal to be recognized correctly.



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Setup & Hold Times

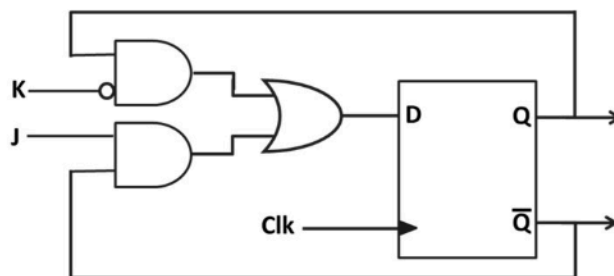
- ♦ A hold time violation is when a input signal change too quickly, after the clock's active transition
- ♦ A setup time violation is when a signal arrives too late, and misses the time when it should advance.
- ♦ If the setup or hold time parameters are violated, the output would be either logic0, logic1 or metastable state (output oscillates between 0 & 1).

Learn this in detail during
advanced course

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FF Conversions



Learn this during advanced
course

$$D = J \overline{Q_n} + \overline{K} Q_n$$

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